



BHAVAN'S PUBLIC SCHOOL, DOHA-QATAR
MIDTERM EXAMINATION MODEL (2026-'27)
MATHEMATICS

CLASS: VI

Time: 3 hours
Maximum Marks: 80

General Instructions:

1. This Question Paper has 4 Sections A, B, C and D.
2. Section A has 19 MCQs and 1 assertion-reason based question of 1 mark each.
3. Section B has 6 questions carrying 02 marks each.
4. Section C has 8 questions carrying 03 marks each.
5. Section D has 6 questions carrying 04 marks each.

SECTION A

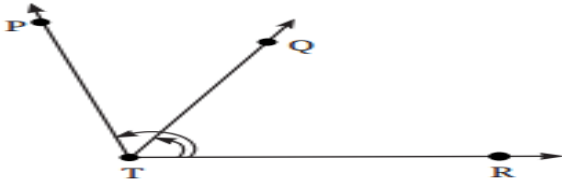
(Multiple Choice Questions- Each question carries 1 mark)

Qn · No		Marks
1	Which of the following is a triangular number? a) 8 b) 6 c) 11 d) 12	1
2	An angle that measures more than 180° but less than 360° is called a _____ angle. a) Acute b) Obtuse c) Reflex d) Straight	1
3	Which of the following has a fixed length? a) Line b) Ray c) Line segment d) Sun rays	1
4	Which graph uses rectangular bars to represent data? a) pictograph b) bar graph c) pie chart d) Frequency distribution table	1
5	The figure formed by two rays with a common endpoint is called an _____. a) line b) line segment c) angle d) circle	1
6	The _____ line in a bar graph represents the X-axis. a) vertical b) horizontal c) slanting d) curved	1
7	Which of the following numbers is divisible by 5? a) 567 b) 205 c) 158 d) 1009	1
8	The smallest 6- digit palindrome number is _____. a) 11111 b) 100000 c) 10100 d) 100001	1

9	Which of the following is a Virahanka number? a) 4 b) 5 c) 7 d) 9	1
10	Which of the following operations is used repeatedly to reach the Kaprekar constant? a) addition b) subtraction c) multiplication d) division	1
11	If two numbers have no common factor other than 1, they are called _____. a) even numbers b) prime numbers c) co-prime numbers d) odd numbers	1
12	Observe the sequence of cube numbers. What is the next number? 1, 8, 27, 64, _____. a) 99 b) 125 c) 101 d) 225	1
13	The tally marks  represents _____. a) 10 b) 15 c) 20 d) 12	1
14	The instrument to measure an angle is a _____. a) ruler b) protractor c) divider d) compasses	1
15	Which of the following is not a palindromic number? a) 121 b) 232 c) 871 d) 454	1
16	If one symbol in a pictograph represents 5 students, then 4 symbols represent _____ students. a) 25 b) 20 c) 50 d) 15	1
17	Which of the following angles represents a full turn? a) 90° b) 180° c) 270° d) 360°	1
18	How many endpoints does a line segment have? a) 2 b) 3 c) 1 d) 0	1
19	Which of the following numbers is both a square number and an even number? a) 6 b) 18 c) 4 d) 10	1
20	<u>Assertion and Reason Question:</u> Assertion (A): 348 is divisible by 2. Reason (R): A number is divisible by 2 if its last digit is 0, 2, 4, 6, or 8. a) Both A and R are true, and R is the correct explanation of A. b) Both A and R are true, but R is not the correct explanation of A. c) A is true, but R is false. d) A is false, but R is true.	1

SECTION – B

This section comprises of very short answer type questions of 2 marks each

21	Using the divisibility rule for 8, determine which of the following numbers are divisible by 8. a) 546840 b) 786111	2
22	Which sequence do you get when you start to add the counting numbers up? Can you give a smaller pictorial explanation?	2
23	Find the common factors of 20 and 28.	2
24	Name the angles marked in the given figure. 	2
25	Find the prime factorization of 456.	2
26	a) How many little squares are there in each shape of the sequence of stacked squares?  b) Which number sequence does this give?	2

SECTION - C

This section comprises of short answer type questions of 3 marks each

27

a) Identify and mark the supercells in the given table.

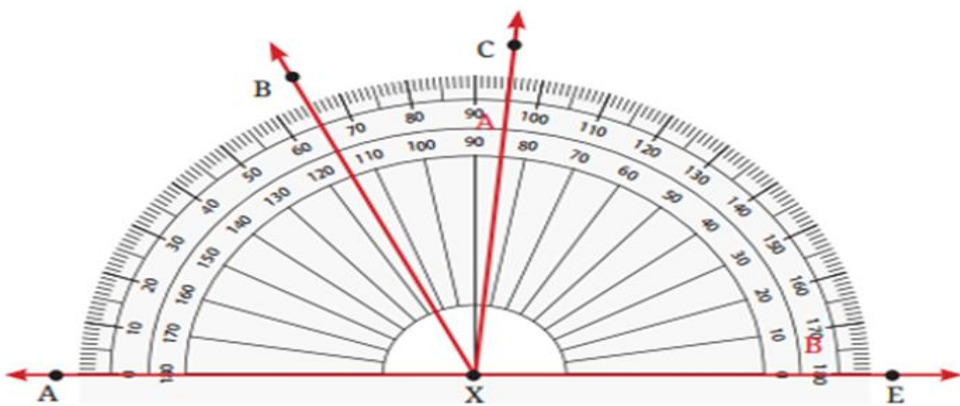
3

987	457	999	123	888	345	789
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b) What would you call the following sequence of numbers?



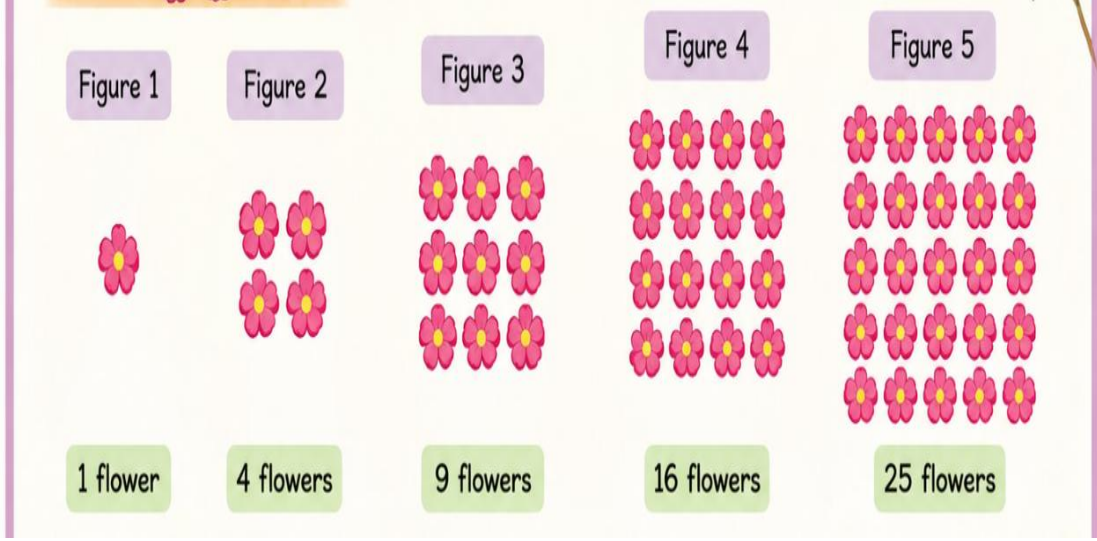
28	During an Environment Day activity, five classes planted different numbers of saplings as shown below. <table><tr><th>Class</th><th>Saplings Planted</th></tr><tr><td>VI A</td><td>12</td></tr><tr><td>VI B</td><td>16</td></tr><tr><td>VI C</td><td>20</td></tr><tr><td>VI D</td><td>24</td></tr><tr><td>VI E</td><td>20</td></tr></table> Draw a bar graph for the given data, using a suitable scale.	Class	Saplings Planted	VI A	12	VI B	16	VI C	20	VI D	24	VI E	20	3
Class	Saplings Planted													
VI A	12													
VI B	16													
VI C	20													
VI D	24													
VI E	20													
29	What is the sum of the smallest and largest 5- digit palindrome? What is their difference?	3												
30	The pictograph represents the number of students participating in different school clubs. Key:  = 5 students <table><tr><th>Club</th><th>Number of Students</th></tr><tr><td>Science Club</td><td>   </td></tr><tr><td>Mathematics Club</td><td>    </td></tr><tr><td>Art Club</td><td>  </td></tr><tr><td>Sports Club</td><td>     </td></tr></table> Answer the following questions: a) How many students are in the Mathematics Club? b) Which club has the highest number of participants? c) How many more students are in the Sports Club than the Art Club? d) Find the total number of students participating in all four clubs.	Club	Number of Students	Science Club	   	Mathematics Club	    	Art Club	  	Sports Club	     	3		
Club	Number of Students													
Science Club	   													
Mathematics Club	    													
Art Club	  													
Sports Club	     													
31	Which of the following pairs of numbers are co-prime? (a) 14 and 25 (b) 22 and 44 (c) 28 and 45	3												

32	The table below shows the number of storybooks read by five students in a month. <table><tr><th>Student</th><th>Number of story books</th></tr><tr><td>Anu</td><td>16</td></tr><tr><td>Diya</td><td>14</td></tr><tr><td>Riya</td><td>12</td></tr><tr><td>Kabir</td><td>10</td></tr><tr><td>Meera</td><td>12</td></tr></table> Prepare a pictograph to represent the given data. Use the key:  = 2 storybooks.	Student	Number of story books	Anu	16	Diya	14	Riya	12	Kabir	10	Meera	12	3
Student	Number of story books													
Anu	16													
Diya	14													
Riya	12													
Kabir	10													
Meera	12													
33	Find the degree measures of $\angle BXE$, $\angle CXE$, $\angle AXB$ and $\angle BXC$ 	3												
34	The marks obtained by 20 students in a Mathematics quiz are: 5, 7, 6, 8, 5, 8, 7, 5, 8, 5, 10, 7, 5, 9, 8, 5, 7, 10, 6, 8 Prepare a frequency distribution table for the given data.	3												

SECTION D

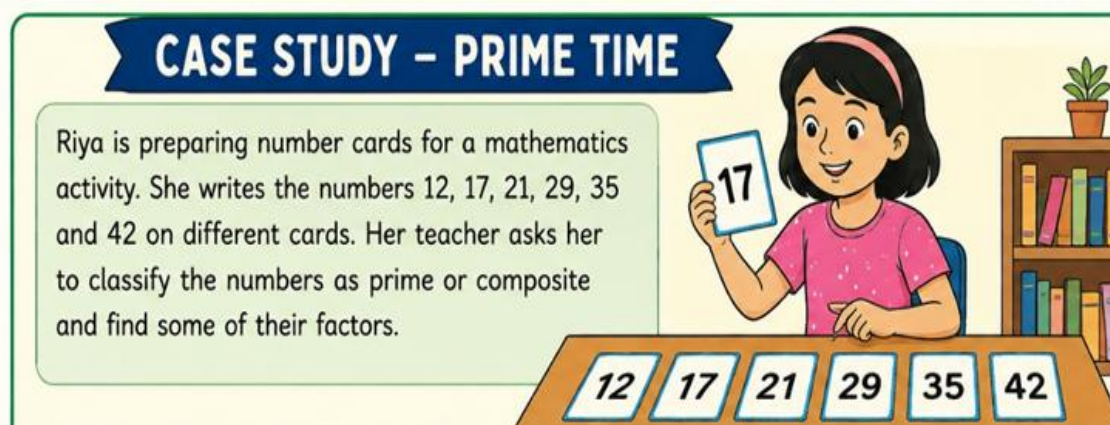
This section comprises of long answer- type questions of 4 marks each

35	Check whether the Collatz Conjecture holds for the starting number 21. Show the sequence of numbers until you reach 1.	4
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36	a) Is 72 divisible by 18? Find the prime factorisations of both 72 and 18. b) Observe the angles given below and identify whether each angle is acute, right, obtuse, or straight. i) 180° ii) 50° iii) 90° iv) 130°	4
37	a) How many rounds does the number 3524 take to reach the Kaprekar constant? b) Is 3366 divisible by 4 ? Why?	4
38	a) Use a protractor to draw angles having the following degree measures: i) 80° ii) 160° b) Write down all the composite numbers from 70 to 80.	4
39	<u>Case Study Question</u> Meera is arranging flowers in the shape of squares. She notices that the number of flowers needed for each figure follows the pattern 1, 4, 9, 16, 25.  Based on the above information, answer the following questions: a. How many flowers will be needed to make Figure 6? b. What type of number pattern is 1, 4, 9, 16, 25? c. Which number in the given pattern is the square of 4? d. Write the next two square numbers after 25.	4

Case Study Question

Riya is preparing number cards for a mathematics activity. She writes the numbers 12, 17, 21, 29, 35 and 42 on different cards. Her teacher asks her to classify the numbers as prime or composite and find some of their factors.



Based on the above information, answer the following questions:

a. Which of the following numbers have exactly two factors?

12 , 17 , 21 , 29 , 35 , 42

b. Which of the following numbers are composite numbers?

12 , 17 , 21 , 29 , 35 , 42

c. Write all the factors of 21.

d. A number has exactly two factors, 1 and itself. What type of number is it?